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REVIEW

of the foreign scientific consultant for the dissertation work of

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“Computer System for Processing and Analysis of Eye Data to Support Laser Coagulation in the Treatment of Diabetic Retinopathy”

submitted for the degree of Doctor of Philosophy (PhD) in the educational program
8D06102 — Computer and Software Engineering

Relevance of the Research Topic

The dissertation addresses a highly relevant problem at the intersection of artificial intelligence, medical image processing, and ophthalmic decision support. Diabetic retinopathy remains an important cause of preventable visual impairment, while effective screening requires the assessment of large numbers of colour fundus images. This challenge is particularly significant in rural and resource-constrained healthcare environments, where specialist capacity may be limited. In this context, the development of reliable and computationally efficient automated methods for diabetic retinopathy assessment is both scientifically and practically important.

A particular strength of the dissertation is that the candidate goes beyond the conventional treatment of image preprocessing as a routine preliminary step. The work advances and systematically investigates the position that preprocessing should be regarded as an integral component of the diagnostic model because it directly influences the representation presented to the convolutional neural network. This perspective is well motivated by the variability of fundus images acquired using different cameras, illumination conditions, and acquisition settings, all of which may contribute to domain shift and reduced generalisation. In my assessment, the research problem is clearly formulated, timely, and appropriately aligned with the scope of Computer and Software Engineering.

Purpose of the Dissertation Research and Scientific Results

The principal aim of the research is to develop and experimentally validate an integrated fundus-image enhancement and CNN-based classification framework for automated multi-stage diabetic retinopathy diagnosis. The dissertation pursues this aim in a logical sequence, beginning with analysis of the problem domain and theoretical foundations, followed by formalisation of the methodology, experimental evaluation, statistical validation, and the design of a screening-system architecture for resource-limited environments.

The proposed methodology incorporates an eight-stage preprocessing pipeline as a defined part of the overall model. The pipeline includes geometric normalisation, field-of-view processing, illumination correction, contrast enhancement, augmentation, and dataset-specific normalisation. Importantly, these components are not merely presented as implementation details; their effects are examined experimentally. The experimental programme is comprehensive. The candidate evaluates the integrated configuration using ResNet-50 and EfficientNet-B3 and compares it with corresponding baseline configurations. The reported results show consistent improvement of the integrated configuration across the primary evaluation criteria. The dissertation further strengthens this analysis through cumulative ablation experiments and parameter-sensitivity studies, allowing the contribution of important preprocessing stages to be examined rather than treating the pipeline as an indivisible black box.

The study also investigates generalisation beyond the primary training corpus. Cross-dataset transfer, external clinical evaluation, device-domain variation, and explainability are considered using several additional datasets. The reported results indicate that the proposed configuration maintains an advantage over the baseline across the evaluated external conditions. The Grad-CAM analysis, complemented by Attention–Lesion Overlap and Intersection-over-Union measurements, provides additional evidence that model attention is more closely associated with expert-annotated lesion regions. Taken together, these experiments provide a broad and technically meaningful evaluation of the proposed framework.

Novelty and Scientific Contribution

In my opinion, the dissertation contains several contributions of doctoral-level methodological significance. The principal contribution is the conceptual and operational reframing of preprocessing as an integral component of a CNN-based diagnostic model and, importantly, the direct experimental evaluation of this proposition under controlled conditions. Within this framework, the candidate develops and evaluates several specific technical elements, including a dual-constraint stochastic CLAHE formulation, adaptive flat-field illumination correction, an explicit field-of-view mask incorporated as an additional input channel, and the Attention–Lesion Overlap metric for quantifying the relationship between model attention and annotated lesions. These elements are integrated into a coherent methodology rather than being presented as isolated algorithmic modifications.

I consider the analysis of domain-distance reduction to be particularly valuable. Instead of assuming that preprocessing improves generalisation, the dissertation attempts to quantify this mechanism by comparing representation-space distances between the source domain and multiple external datasets. This transforms a commonly stated explanation into an experimentally testable proposition. The treatment of model initialisation is also methodologically sound: alternative pretraining strategies are evaluated using a defined acceptance procedure, and a negative result for label-free self-supervision trained from scratch is reported transparently. This willingness to report evidence that does not support an initially plausible approach adds credibility to the research.

Degree of Reliability and Validity of the Scientific Results

The reliability of the reported findings is well supported by the breadth of the empirical evaluation and the statistical framework adopted in the dissertation. The research draws on eight datasets serving different experimental purposes, including primary training and ablation, cross-dataset transfer, external clinical evaluation, explainability, device-domain analysis, and evaluation in a data-scarce setting. This diversity is a significant strength because it reduces reliance on conclusions drawn from a single dataset or acquisition environment. The experimental design employs five-fold patient-level stratified cross-validation with leakage control and evaluates performance using multiple complementary measures, including weighted F1, ROC-AUC, quadratically weighted Cohen's Kappa, and accuracy. Statistical assessment includes McNemar and DeLong tests, mixed-effects modelling, bootstrap confidence intervals, and corrections for multiple comparisons.

I particularly appreciate that the dissertation defines success criteria for the principal hypotheses in advance and explicitly discusses the boundary conditions and limitations of the evidence. The candidate generally avoids presenting experimental findings as evidence of clinical deployment readiness and distinguishes demonstrated results from proposed system-level designs. This careful interpretation reflects appropriate scientific maturity and strengthens confidence in the conclusions.

Practical Significance of the Results

The work has clear potential practical relevance. The preprocessing pipeline is specified stage by stage, including its operators, parameters, and sequence, and the implementation is provided in an appendix. This level of specification supports reproducibility and facilitates further evaluation by other researchers. The dissertation also proposes a modular architecture for automated diabetic retinopathy screening that includes a configurable preprocessing engine, an inference module, telemedicine support, and a physician-in-the-loop decision-support interface. The design considers integration with PACS, EHR systems, and national eHealth infrastructure and is oriented towards rural and underserved settings, including the healthcare context of Kazakhstan.

It is appropriate that the proposed architecture retains the clinician as the final decision-maker. While the architecture remains a design specification rather than a clinically deployed system, it provides a credible framework for subsequent prototype development, prospective validation, and assessment in real healthcare environments. Its emphasis on computational constraints and interoperability further enhances the practical relevance of the research.

Academic Contribution and Doctoral Qualification

Based on my evaluation of the dissertation, the candidate demonstrates the level of technical competence and research maturity expected of doctoral work. The dissertation shows a strong command of deep learning, image-processing methods, experimental design, and statistical validation. The research problem is clearly identified, the methodology is systematically developed, and the experimental programme is sufficiently broad to support the principal conclusions within the stated limitations. The work is also characterised by appropriate scientific transparency. Positive findings are supported

through multiple forms of evaluation, while limitations, negative findings, and boundaries of interpretation are explicitly acknowledged. The research has additionally resulted in peer-reviewed publications and international conference dissemination, indicating engagement with the wider scientific community.

Overall, I consider this dissertation to be scientifically sound, methodologically rigorous, technically substantial, and relevant to contemporary research in artificial intelligence and medical image analysis. The study makes a meaningful contribution to automated diabetic retinopathy analysis, particularly through its systematic investigation of preprocessing as an integral component of the diagnostic framework. I therefore give a positive evaluation of the dissertation and recommend it for defence. In my opinion, the dissertation satisfies the academic and scientific requirements for the award of the degree of Doctor of Philosophy (PhD) in the educational program 8D06102 — Computer and Software Engineering.



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